

Field Demonstration Videos Technical Manual

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1. Index of 11 Practical Field Demonstration Videos

VID-01: Induction Motor Star-Delta Starting Transition & Contactors
VID-02: Motor Insulation Resistance (Megger) Testing (500V/1000V DC)
VID-03: Motor Terminal Box Bridging (Star vs Delta Configuration)
VID-04: Motor Power Factor Capacitor Bank Connection & Sizing
VID-05: Transformer Insulation & Polarization Index (PI $\geq$ 2.0)
VID-06: Transformer Turns Ratio (TTR) Testing & Verification
VID-07: Transformer Off-Circuit Tap Changer Step Adjustment ($\pm$ 2.5%)
VID-08: Digital Multimeter True-RMS Voltage & Current Measurements
VID-09: Phase Rotation Sequence Meter Verification (Clockwise A-B-C)
VID-10: Earth Ground Resistance 3-Point Fall-of-Potential (62% Rule)
VID-11: Infrared Thermography Inspection of Switchgear & Breakers

2. Key Safety & Diagnostic Guidelines

1. Always verify zero voltage with an approved two-pole tester before touching.
2. Discharge capacitive residual charges on transformers and cables after Megger testing.
3. Never operate off-circuit tap changers while transformer is energized.
4. Ensure 30-50 ms dead-time between Star and Delta contactor switching.